Numerical Partial Differential Equation

Exercise

The following partial differential equation describes a vibrating string,

$$\frac{\partial^2 u}{\partial t^2} = \frac{1}{4}\frac{\partial^2 u}{\partial x^2}, \text{for } 0 \le x \le 0.5, t > 0.$$

The initial condition $u(x,0) = 0$ and $\frac{\partial u}{\partial t}(x,0) = \sin 4\pi x$ for $0 \le x \le 0.5$. The boundary condition with $u(0,t) = u(0.5,t) = 0$, for $t \ge 0$.

a) What is the category of the partial differential equation given? Justify your answer.

b) Use $\Delta t = \Delta x = 0.1$ to solve the propagation of the vibrating string up $t = 0.1$.

Exercise

Solution:

Exercise2

The partial differential equation of heat conservation on a rod with length of $2cm$ and time, $t > 0s$ given as,

$$\frac{\partial u}{\partial t} = \frac{1}{10}\frac{\partial^2 u}{\partial x^2}, \qquad \text{for } 0 < x < 1, t > 0$$

Initially, at $t = 0$ the condition is expressed as

$$u(x, 0) = \sin\left(\frac{\pi}{2}x\right),$$

for all positions of the rod between $0 \leq x \leq 2$. The condition at both ends of the rod is equal to zero for $t \geq 0$, which is expressed as

$$u(0, t) = u(2, t) = 0.$$

a) Categorise the partial differential equation above.

b) Evaluate the heat conservation across the rod for every $5mm$ at $t = 1s$ and $t = 2s$ with the aid of graphical grids.

Exercise2

Solution: